MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 1

Notation 1. We will use *summation notation*: The symbol

$$\sum_{k=1}^{n} a_k$$

has to be read like a for loop: The entries $a_1, a_2, \ldots, a_n$ are numbers that are indexed by their subscripts, and the symbol above is short for the iterated sum

$$a_1 + a_2 + \cdots + a_n.$$

(1) Suppose that we have $a, b \in \mathbb{Z}$.

(a) If a, b each leave a remainder of 1 after dividing by 4, what can you say about the remainder that ab leaves when divided by 4?

Solution. Our assumption translates concretely to saying that there exist integers $k, l \in \mathbb{Z}$ with $a = 4k + 1$ and $b = 4l + 1$. The product is

$$ab = (4k+1)(4l+1) = 16kl + 4l + 4k + 1 = 4(4kl + l + k) + 1,$$

which clearly has remainder 1 when divided by 4.

(b) If a, b each leave a remainder of 3 after dividing by 4, what can you say about the remainder that ab leaves when divided by 4?

Solution. Just as before, except that we have $a = 4k + 3$ and $b = 4l + 3$ with the product given by

$$ab = (4k+3)(4l+3) = 16kl + 12k + 12l + 9 = 4(4kl + 3k + 3l + 2) + 1,$$

which has remainder 1 when divided by 4.

Remark 1. In the notation introduced in Lecture 4, we can encapsulate this as follows:

(a) If $a \equiv 1 \pmod 4$ and $b \equiv 1 \pmod 4$, then $ab \equiv 1 \pmod 4$.

(b) If $a \equiv 3 \pmod 4$ and $b \equiv 3 \pmod 4$, then $ab \equiv 1 \pmod 4$.

These are of course reasonable things to be true!

(2) Use the previous problem to show that there are infinitely many primes (such as $3, 7, 11, 19, \ldots$) that leave a remainder of 3 after dividing by 4.

Hint: This uses a variation of the construction used in Euclid's proof of the infinitude of all primes: If $p_1, \ldots, p_r$ is a finite set of primes, consider the prime factorization of $N = 4p_1 \cdots p_r - 1$.

Solution. Let X be the set of primes that leave a remainder of 3 after dividing by 4. We will show that, for any finite subset $\{p_1, \ldots, p_r\} \subset X$, we can find $q \in X$ such that q does not belong to $\{p_1, \ldots, p_r\}$. This is the concrete manifestation of the infinitude of X.

To do this, as the hint says, we will consider the number

$$N = 4p_1 \cdots p_r - 1.$$

As we argued in Lecture 2, this natural number is a product of prime numbers: $N = q_1 \cdots q_m$. Furthermore, we know the following things:

- N is odd (why?) meaning none of the primes q_j can be 2;
- None of the primes $q_1, \ldots, q_m$ belongs to the finite set $\{p_1, \ldots, p_r\}$: this is because, *by its construction*, none of the primes in this finite set can divide N (why?);
- N leaves a remainder of 3 when divided by 4. In the notation from Lecture 4, we have

$$N \equiv 3 \pmod 4.$$

Now, the first point means that each q_i is congruent to either 1 or 3 mod 4 (why?). If *all* of them happened to be 1 mod 4, then problem 1 shows that their product (which is N) would also be 1 mod 4. But the last point tells us that N is congruent to 3 mod 4. The only way for this to be possible is if at least *one* of $q_1, \ldots, q_m$ is congruent to 3 mod 4. Up to reindexing the subscripts, we can assume that $q_1 \equiv 3 \pmod 4$. Now, the second bullet point tells us that $q = q_1$ is a *new* element of X that's not one of the already known $p_1, \ldots, p_r$.

This completes the proof.

(3) Prove the following formulas using induction:

(a)

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6};$$

Solution. **Base case**: When $n = 1$, the left hand side is $1^2 = 1$ and the right hand side is $\frac{1(1+1)(2+1)}{6} = \frac{2 \cdot 3}{6} = 1$. These are clearly equal.

Inductive hypothesis: We have an equality

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

for n.

Inductive step: We need to verify the equality with n replaced by $n+1$. For this we add $(n+1)^2$ to both sides of our inductive hypothesis. This gives us

$$\sum_{k=1}^{n+1} k^2$$

on the left hand side, and

$$\frac{n(n+1)(2n+1)}{6} + (n+1)^2$$

on the right. Let us manipulate the RHS:

$$\begin{aligned} \frac{n(n+1)(2n+1)}{6} + (n+1)^2 &= \frac{n(n+1)(2n+1) + 6(n+1)^2}{6} \\ &= \frac{(n+1)\left(n(2n+1) + 6(n+1)\right)}{6} \\ &= \frac{(n+1)\left(2n^2 + n + 6n + 6\right)}{6} \\ &= \frac{(n+1)(2n^2 + 7n + 6)}{6}. \end{aligned}$$

We would like for this to be equal to:

$$\begin{aligned} \frac{(n+1)(n+2)(2(n+1)+1)}{6} &= \frac{(n+1)(n+2)(2n+2+1)}{6} \\ &= \frac{(n+1)(n+2)(2n+3)}{6} \\ &= \frac{(n+1)(2n^2 + 7n + 6)}{6}. \end{aligned}$$

The two are now clearly the same thing!

(b)

$$\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}.$$

Solution. Same as the previous problem. The inductive step comes down to the algebraic manipulations

$$\begin{aligned}\frac{n^2(n+1)^2}{4} + (n+1)^3 &= \frac{n^2(n+1)^2 + 4(n+1)^3}{4} \\ &= \frac{(n+1)^2(n^2+4(n+1))}{4} \\ &= \frac{(n+1)^2(n^2+4n+4)}{4} \\ &= \frac{(n+1)^2(n+2)^2}{4}.\end{aligned}$$

(4) Prove that $2^{2n+1}+1$ is a multiple of 3 for every $n \in \mathbb{N}$.

Solution. **Base case**: When $n=1$, this is $2^3+1=9$, which is $3 \cdot 3$.
Inductive hypothesis: $2^{2n+1}+1=3k$ for some $k \in \mathbb{N}$.
Inductive step: Our inductive hypothesis can be rewritten as

$$2^{2n+1} = 3k-1.$$

Now, if we plug in $n+1$ for n, we get

$$2^{2(n+1)+1}+1 = 2^{2n+3}+1 = 4 \cdot 2^{2n+1}+1 = 4(3k-1)+1 = 12k-4+1 = 12k+3,$$

and this is of course a multiple of 3.

Remark 2. We can also prove this directly using mod-3 arithmetic. We have

$$2 \equiv -1 \pmod 3 \Rightarrow 2^{2n+1} \equiv (-1)^{2n+1} \equiv -1 \pmod 3.$$

So adding 1 to this gives us $0 \pmod 3$, and so shows that $2^{2n+1}+1$ is a multiple of 3.

(5) Prove that for every $n \in \mathbb{N}$, we have

$$\sum_{k=1}^{n} \frac{1}{k^2} \leq 2 - \frac{1}{n}.$$

Solution. **Base case**: When $n=1$, the LHS is just $1/1^2=1$, while the RHS is $2-\frac{1}{1}=1$, and we clearly have $1 \leq 1$.
Inductive hypothesis:

$$\sum_{k=1}^{n} \frac{1}{k^2} \leq 2 - \frac{1}{n}.$$

Inductive step: Add $\frac{1}{(n+1)^2}$ to both sides to get the inequality

$$\sum_{k=1}^{n+1} \frac{1}{k^2} \leq 2 - \frac{1}{n} + \frac{1}{(n+1)^2}.$$

To finish, we must show that

$$2 - \frac{1}{n} + \frac{1}{(n+1)^2} \leq 2 - \frac{1}{n+1}.$$

This would follow if we knew:

$$\frac{1}{(n+1)^2} \leq \frac{1}{n} - \frac{1}{n+1}.$$

The RHS here is

$$\frac{1}{n} - \frac{1}{n+1} = \frac{n+1-n}{n(n+1)} = \frac{1}{n(n+1)}$$

Now, the desired inequality
$$\frac{1}{(n+1)^2} \leq \frac{1}{n(n+1)}$$
is true because the denominator on the LHS is clearly larger than the one on the RHS.

(6) Use l'Hôpital's rule and induction to prove that
$$\lim_{x\to\infty} \frac{x^n}{e^x} = 0$$
for all $n \in \mathbb{N}$. You may assume that the equality is true when $n = 0$.

Solution. **Base case**: Though the problem asks you to show this for all $n \in \mathbb{N}$, there is actually only additional generality in showing it for all *non-negative* integers, including 0. So we will start the base case at $n = 0$, where we are told we can assume the truth of the equality.

Inductive hypothesis:
$$\lim_{x\to\infty} \frac{x^n}{e^x} = 0$$

Inductive step:

Note that x^{n+1} and e^x are going to ∞ as $x \to \infty$. Therefore, l'Hôpital's rule applies and tells us that we have
$$\lim_{x\to\infty} \frac{x^{n+1}}{e^x} = \lim_{x\to\infty} \frac{(n+1)x^n}{e^x} = (n+1) \lim_{x\to\infty} \frac{x^n}{e^x} = (n+1) \cdot 0 = 0.$$
In the penultimate equality, we have used our inductive hypothesis.